

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0826

West Paintsville 138 kV
December 01, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0826. The Interconnection Customer proposes a 111.6 MW solar photovoltaic facility interconnecting at the West Paintsville 138 kV substation in the Kentucky area. The request was received on July 13, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,924,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dowell to West Paintsville 138.0 kV	Wissota to Nsp 138.0 kV	118.8	312	26.9
Wichita Plant to Tinkertown 138.0 kV	Davisville to N Appleton Substation 138.0 kV	117.7	190	8.2
Tinkertown to Stillwater B 138.0 kV	Davisville to N Appleton 138.0 kV	116.2	208	8.2
Beck Road to Sporn Substation 230.0 kV	Beck Rd to Sporn 345.0 kV	105.9	508	5.6
Topmost to Stillwater B 138.0 kV Line	Davisville to N Appleton 138.0 kV	105.6	213	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dowell Substation to W Paintsville 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,623,000	14% (\$361,000)
Wichita Plant to Tinkertown 138.0 kV	Rebuild line with higher-capacity conductor	\$2,206,000	8% (\$179,000)
Tinkertown to Stillwater B 138.0 kV	Rebuild line with higher-capacity conductor	\$6,614,000	8% (\$537,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beck Road to Sporn Substation 230.0 kV	Reconductor line	\$1,686,000	2% (\$36,000)
Topmost to Stillwater B Substation 138.0 kV Line	Reconductor line	\$9,986,000	8% (\$811,000)
	Total allocated to project		\$1,924,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.